attempt (a wormhole, a spinning cylinder,[13] *a "cosmic string"*[14] *or whatever), just before one's device becomes a time machine, a beam of vacuum fluctuations will circulate through the device and destroy it* Hawking seems ready to bet heavily on this outcome.

I am *not* willing to take the other side in such a bet. I *do* enjoy making bets with Hawking, but only bets that I have a reasonable chance of winning. My strong gut feeling is that I would lose this one. My own calculations with Kim, and unpublished calculations that Eanna Flanagan (a student of mine) has done more recently, suggest to me that Hawking is likely to be right. ***Every time machine is likely to self destruct (by means of circulating vacuum fluctuations) at the moment one tries to activate it.*** However, we cannot know for sure until physicists have fathomed in depth the laws of quantum gravity.

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1. I have chosen to write this chapter solely from my own personal viewpoint. It therefore is much less objective than the rest of the book, and represents other people's research much less fairly and less completely than it does my own.

2. See the "Best Guesses" section of Chapter 13.

3. In technical language, we say that the exotic material "violates the averaged weak energy condition."

4. See especially pages 347, 348, and 406 of *Contact* by Carl Sagan. There the exotic condition (negative average energy density as seen by light beams traveling through the wormhole) is expressed in a different, but equivalent way: As seen by someone at rest inside the wormhole, the material must have a large tension, along the radial direction, a tension that is bigger than the material's energy density.

5. Chapter 13.

6. The Planck–Wheeler length is the square root of the Planck–Wheeler area (which entered into the formula for the entropy of a black hole, Chapter 12); it is given by the formula $\sqrt{Gh/c^3}$, where G = 6.670×10^{-8} dyne-centimeter^2/gram^2 is Newton's gravitation constant, $\hbar = 1.055 \times 10^{-27}$ erg-second is Planck's quantum mechanical constant, and $c = 2.998 \times 10^{10}$ centimeter/second is the speed of light.

7. I wish that I could draw a simple, clear picture to show how this smooth creation of a wormhole is accomplished; unfortunately, I cannot.

8. This time machine and others described later in this chapter are by no means the first time machine—type solutions to the Einstein field equation that people have found. In 1937, W. J. van Stockum in Edinburgh discovered a solution in which an infinitely long, rapidly spinning cylinder functions as a time machine. Physicists have long objected that nothing in the Universe can be infinitely long, and they have suspected (but nobody has proved) that, if the length of the cylinder were made finite, it would cease to be a time machine. In 1949, Kurt Godel, at the Institute for Advanced Study in Princeton, New Jersey, found a solution to Einstein's equation that describes a whole universe which